COMPLETE ALGEBRAS AND INVERSE SYSTEMS ARE EQUIVALENT, AND EVERY COMPLETE OBJECT IS A QUOTIENT OF A COMPLETED POLYNOMIAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof verifies the quasi-inverse property through the limit lemma, checks the completed polynomial ring is complete, and exhibits an arbitrary complete object as a quotient by extending a mod-I surjection and applying the generalities lemma on completion

Editorial note. The label lemma-topologically-finite-type names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the categories C of inverse systems (B_n) of discrete A/I^n-modules with the compatibility conditions and C' of I-adically complete A-algebras, with the functors between them (equations of the same section)
- the limit lemma: B = lim B_n is I-adically complete with B/I^nB = B_n (Algebra, Lemma algebra-lemma-limit-complete)
- completeness of the completed polynomial ring A[x_1,...,x_r]^wedge = lim A_n[x_1,...,x_r]
- the surjectivity criterion for maps into complete rings: a map surjective modulo I into a complete target is surjective (Algebra, Lemma algebra-lemma-completion-generalities)

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Prerequisite pointer invoked at proof line(s) 382; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09B8: sources/stacks-chow/tags.txt, source lines 11786–11786. Prerequisite pointer invoked at proof line(s) 364; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AJN: sources/stacks-restricted/pinned-original-source.tex, source lines 337–337. Prerequisite pointer invoked at proof line(s) 353; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a ring and let $I \subset A$ be a finitely generated ideal. The functor*

$$\mathcal{C} \longrightarrow \mathcal{C}', \quad (B_n) \longmapsto B = \lim B_n$$

is a quasi-inverse to (0AJN). The completions $A[x_1, \ldots, x_r]^\wedge$ are in $\mathcal{C}'$ and any object of $\mathcal{C}'$ is of the form

$$B = A[x_1, \ldots, x_r]^\wedge / J$$

for some ideal $J \subset A[x_1, \ldots, x_r]^\wedge$.

Proof. Let (B_n) be an object of $\mathcal{C}$. By Algebra, Lemma 09B8 we see that $B = \lim B_n$ is I-adically complete and $B/I^nB = B_n$. Hence we see that B is an object of $\mathcal{C}'$ and that we can recover the object (B_n) by taking the quotients. Conversely, if B is an object of $\mathcal{C}'$, then $B = \lim B/I^nB$ by assumption. Thus $B \mapsto (B/I^nB)$ is a quasi-inverse to the functor of the lemma.

Since $A[x_1, \ldots, x_r]^\wedge = \lim A_n[x_1, \ldots, x_r]$ it is an object of $\mathcal{C}'$ by the first statement of the lemma. Finally, let B be an object of $\mathcal{C}'$. Choose $b_1, \ldots, b_r \in B$ whose images in B/IB generate B/IB as an algebra over A/I. Since B is I-adically complete, the A-algebra map $A[x_1, \ldots, x_r] \to B$, $x_i \mapsto b_i$ extends to an A-algebra map $A[x_1, \ldots, x_r]^\wedge \to B$. To finish the proof we have to show this map is surjective which follows from Algebra, Lemma 0315 as our map $A[x_1, \ldots, x_r] \to B$ is surjective modulo I and as $B = B^\wedge$. $\square$